

REAL TIME WIREFRAME-BASED POTHOLE DETECTION AND AVOIDANCE SYSTEM

BATCH - 02

GUIDED BY: DR RG SURESH KUMAR

PRESENTED BY: MOHAMED B SIRAJUDDEEN

AGENDA

- | | | | |
|---|-----------------------|----|-----------------------------------|
| 1 | ABSTRACT | 9 | PROPOSED SYSTEM |
| 2 | DOMAIN | 10 | PROPOSED ARCHITECTURE |
| 3 | EXISTING WORK | 11 | H/W REQUIREMENT & S/W REQUIREMENT |
| 4 | EXISTING ARCHITECTURE | 12 | ENVIRONMENT SETUP SCREEN SHOT |
| 5 | PROBLEM STATEMENT | 13 | IMPLEMENTATION DETAILS |
| 6 | LITERATURE SURVEY | 14 | RESULT & DISCUSSION |
| 7 | SURVEY SUMMARY TABLE | 15 | CONCLUSION & FUTURE ENHANCEMENT |
| 8 | PROPOSED SOLUTION | 16 | REFERENCE |

ABSTRACT

Road surface irregularities like potholes present major risks to vehicle safety, passenger comfort, and the efficiency of traffic flow. These issues have significant implications in the transportation domain, as they directly affect road safety, traffic management, and overall driving experience. With the growth of autonomous vehicles and smart city initiatives, the need for intelligent systems that can accurately detect and analyze road conditions in real time has become increasingly important.

Traditional approaches for pothole detection have been enhanced through the integration of computer vision and artificial intelligence. These systems utilize cameras to capture road images, which are then processed using deep learning models to detect and classify potholes. By applying AI-based image analysis, the existing methods effectively automate the identification of road surface irregularities, offering a faster and more efficient alternative to manual inspections and sensor-based techniques.

The limitation in existing systems is their reliance on two-dimensional bounding boxes or simple classification, which lack depth and shape information. To address this limitation, my project proposes a Real-Time Wireframe Pothole Detection System that generates a three-dimensional wireframe of road surfaces using camera input. The proposed model aims to integrate depth estimation with wireframe modeling for precise analysis of surface irregularities, enabling better decisions for autonomous vehicles and improving road safety.

DOMAIN AND SUB-DOMAIN

ARTIFICIAL INTELLIGENCE

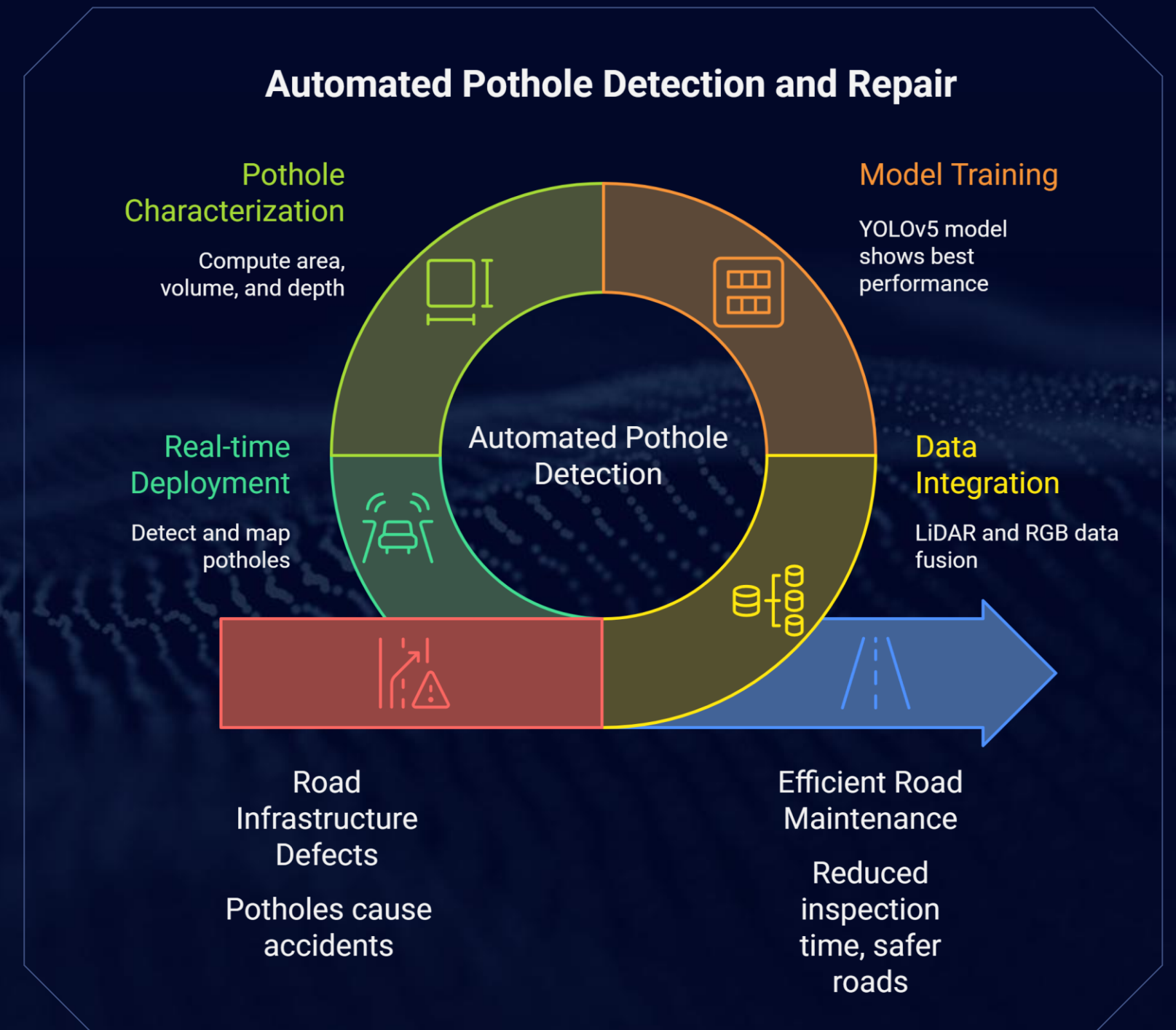
Encompasses computational models that replicate human cognitive functions such as perception, learning, reasoning, and autonomous decision-making.

AUTONOMOUS VEHICLES

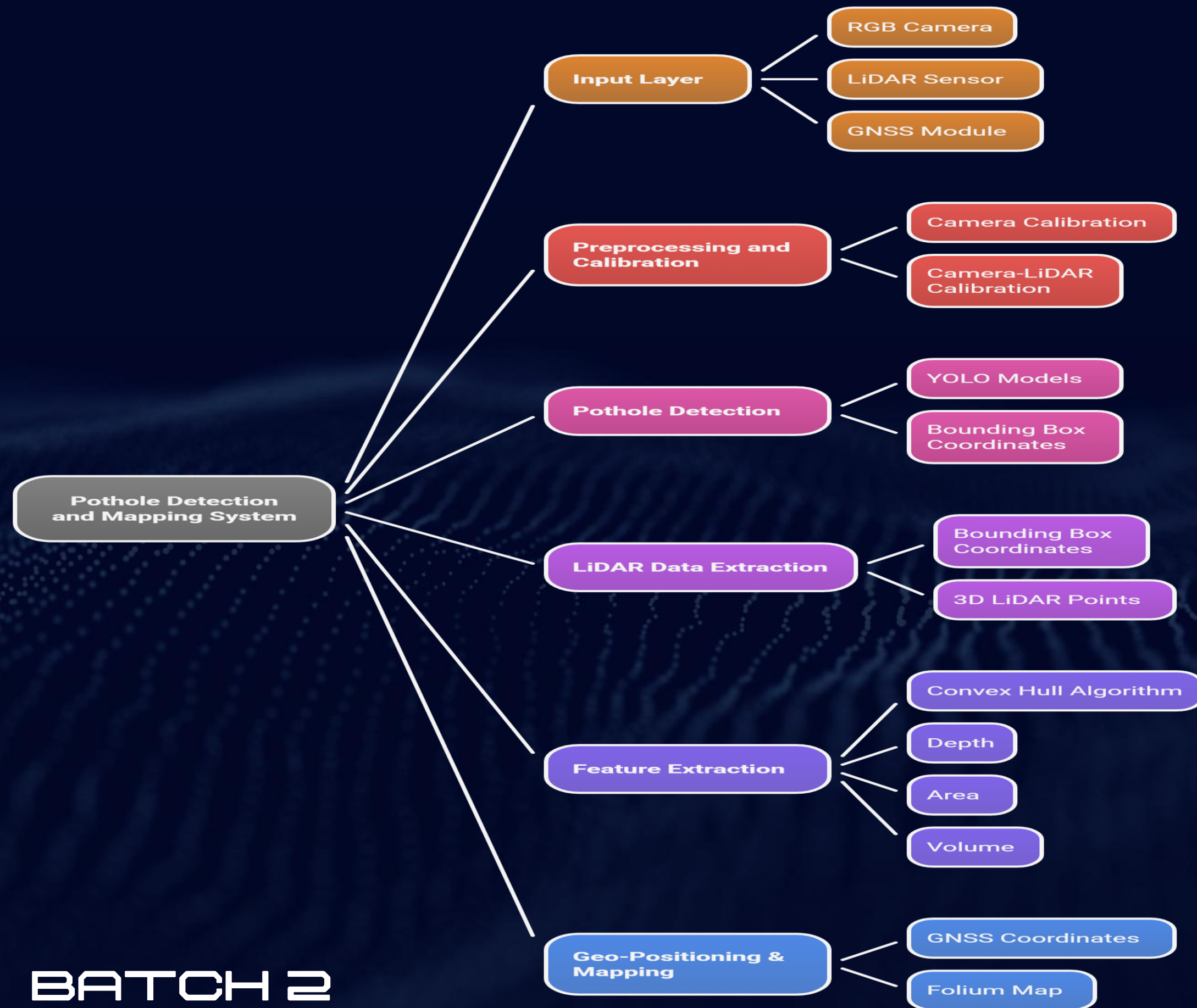
Focuses on AI-driven perception and control in autonomous vehicles through deep learning, monocular depth estimation, and real-time 3D scene understanding for safe navigation and environment interaction.

EXISTING WORK

- ❖ LiDAR + Camera fusion for accurate depth
- ❖ YOLO object detection
- ❖ 3D LiDAR depth analysis
- ❖ GNSS-based geo-positioning to automatically detect, measure, and map potholes.
- ❖ This eliminates manual road inspection and improves both accuracy and maintenance efficiency.
- ❖ Generates 3D point cloud data for area, volume, and depth analysis
- ❖ Point cloud and Convex Hull for geometry



EXISTING ARCHITECTURE



PROBLEM STATEMENT

- 1. Economic Impracticality:** LiDAR-based pothole detection systems are prohibitively expensive for mass-scale autonomous vehicle deployment, with high unit costs, complex calibration pipelines, and heavy computational requirements making them economically unsustainable.
- 2. Physical and Environmental Limitations:** LiDAR fails to reliably detect water-filled potholes due to laser scattering caused by water's refractive index, and its bulky external mounting disrupts vehicle aerodynamics — making it structurally impractical for all types of real-world vehicle deployment.

PAPER 1

TITLE: An Enhanced YOLOv8 Model for Real-Time and Accurate Pothole Detection and Measurement

INTRODUCTION:

Potholes are a major form of road surface damage caused by water infiltration, heavy traffic, and weather conditions, posing serious risks to vehicles and road users. Traditional manual inspection methods are slow, costly, and prone to human error. To overcome these limitations, **AI- and vision-based detection systems** offer faster, more accurate, and cost-effective solutions. Technologies such as **LiDAR, stereo cameras, and deep learning models** enable automated pothole identification and measurement, improving maintenance efficiency and road safety.

CONCLUSION:

They created the PothRGBD RGB-D dataset and proposed a lightweight YOLOv8n-seg variant that improves segmentation and enables accurate perimeter & depth estimation. Therefore, I can conclude from their paper that integrating DSConv and SimAM into a YOLOv8 segmentation yields more accurate pothole masks and reliable physical measurements, which I can apply to my pothole detection project to get precise segmentation masks and direct depth/perimeter estimates.

PAPER 2

TITLE: Robust Video Based Pothole Detection and Area Estimation for Intelligent Vehicles with Depth Map and Kalman Smoothing

INTRODUCTION:

Autonomous driving systems require precise road perception for safety and comfort, but potholes remain a major challenge. Existing LiDAR-based methods are accurate but costly, while vision-only approaches struggle with lighting and angle variations. To address this, a **robust vision-based framework** is proposed using **ACSH-YOLOv8** for detection, **DepthAnything V2** for depth estimation, and a **Minimum Bounding Triangulated Pixel (MBTP)** method for accurate area measurement. A **CDKF-based Kalman Filter** enhances stability across video frames, ensuring reliable pothole detection and area estimation in real-world conditions.

CONCLUSION:

They present an end-to-end vision pipeline to achieve accurate and stable pothole area estimation from video frames. I conclude from their paper that integrating enhanced object detection, monocular depth estimation, pixel-level area computation, and Kalman-based smoothing can significantly improve the precision and reliability of pothole detection.

PAPER 3

TITLE: Mobile3DRecon: Real-time Monocular 3D Reconstruction on a Mobile Phone

INTRODUCTION:

Augmented Reality (AR) requires accurate 3D reconstruction to achieve realistic integration of virtual objects with real environments. Existing systems like Kinectic Fusion and Bundle Fusion depend on RGB-D cameras and high computational power, making them unsuitable for mid-range mobile devices. To address this, the **Mobile3DRecon** system enables **real-time dense 3D surface reconstruction** using only a **monocular camera**, without extra sensors. It introduces **multi-view depth estimation** and **incremental mesh generation**, allowing seamless, real-time AR performance on standard mobile platforms.

CONCLUSION:

They presented a mobile system capable of generating dense, real time 3D surface meshes using only an embedded monocular camera. They show robust results for AR and object interaction on mid-range phones. Therefore, I can conclude that their mobile-focused, CPU-based incremental mesh generation offers techniques and system design ideas directly applicable to resource-constrained 3D reconstruction tasks like pothole mapping or detection.

PAPER 4

TITLE: Pothole and Plain Road Classification Using Adaptive Mutation Dipper Throated Optimization and Transfer Learning for Self-Driving Cars.

INTRODUCTION:

Autonomous driving relies on accurate environmental perception to ensure safety, with **pothole detection** being a critical component. Traditional sensor- and vision-based methods face challenges such as high cost, computational complexity, and limited accuracy under varying conditions. To overcome these, this study proposes a **deep learning–based pothole detection system** using a **convolutional neural network (CNN)** for feature extraction and a **Random Forest classifier** optimized through **Adaptive Mutation Dipper Throated Optimization (AMDTO)**. This approach enhances detection accuracy, robustness, and real-time performance for self-driving vehicles.

CONCLUSION:

This study showed a pipeline that combines transfer-learning feature extraction. Therefore, I can conclude from their paper that applying metaheuristic optimization for feature selection and classifier tuning together with an improved SMOTE balancing step can dramatically boost pothole vs plain-road classification accuracy.

P A P E R S

TITLE: Real-time scene reconstruction and triangle mesh generation using multiple RGB-D cameras

INTRODUCTION:

3D surface reconstruction and meshing have been extensively studied in computer vision and graphics, with applications in fields like archaeology, cinematography, and robotics. While most work focuses on static scenes, dynamic environments—required in applications like telepresence—pose additional challenges. 3D reconstruction typically involves three stages: data acquisition, surface reconstruction, and geometry extraction. RGB-D sensors enable real-time reconstruction but introduce noise and limited coverage. This work enhances a moving least squares (MLS) surface reconstruction method and generates triangle meshes, a common and view-independent representation in graphics.

CONCLUSION:

They proposed a real-time 3D reconstruction method with a GPU-optimized mesh merging method. They demonstrate both high mesh quality and significant acceleration compared to CPU based methods. I can conclude that their GPU parallel mesh approach, especially the MLS point cloud smoothing and mesh merging, gives insight into achieving fast, reliable surface modeling— knowledge that is valuable for enhancing the efficiency of mesh processing in pothole detection systems.

SURVEY SUMMARY TABLE

SL. NO	TITLE	YEAR OF PUBLISHING	TECHNOLOGY USED	CONCLUSION
1	An Enhanced YOLOv8 Model for Real-Time and Accurate Pothole Detection and Measurement.	2025	YOLOv8n-seg, Dynamic Snake Convolution (DSConv), Simple Attention Module (SimAM), GELU activation.	They created the PothRGBD RGB-D dataset and proposed a lightweight YOLOv8n-seg variant that improves segmentation and enables accurate perimeter & depth estimation. Therefore, I can conclude from their paper that integrating DSConv and SimAM into a YOLOv8 segmentation yields more accurate pothole masks and reliable physical measurements, which I can apply to my pothole-detection project to get precise segmentation masks and direct depth/perimeter estimates.
2	Robust Video-Based Pothole Detection and Area Estimation for Intelligent Vehicles with Depth Map and Kalman Smoothing	2025	ACSH-YOLOv8, BoT-SORT, Ego-motion compensation, DepthAnything, MBTP, CDKF (Confidence–Distance Kalman Filter).	They present an end-to-end vision pipeline to achieve accurate and stable pothole area estimation from video frames. I conclude from their paper that integrating enhanced object detection, monocular depth estimation, pixel-level area computation, and Kalman-based smoothing can significantly improve the precision and reliability of pothole detection.
3	Mobile3DRecon: Real-time Monocular 3D Reconstruction on a Mobile Phone	2020	Monocular camera, 6DoF SLAM system, Semi-global matching (SGM), DNN-based depth refinement, Scalable TSDF voxel fusion, Mesh generation algorithm on mobile CPUs.	They presented a mobile system capable of generating dense, real-time 3D surface meshes using only an embedded monocular camera. They show robust results for AR and object interaction on mid-range phones. Therefore, I can conclude that their mobile-focused, CPU-based incremental mesh generation offers techniques and system design ideas directly applicable to resource-constrained 3D reconstruction tasks like pothole mapping or detection.

SURVEY SUMMARY TABLE

SL. NO	TITLE	YEAR OF PUBLISHING	TECHNOLOGY USED	CONCLUSION
4	Pothole and Plain Road Classification Using Adaptive Mutation Dipper Throated Optimization and Transfer Learning for Self-Driving Cars.	2022	ResNet-50, Binary Advanced Mutation, Dipper Throated Optimization, Optimized hashing SMOTE, Random Forest (RF), AMDTO optimizer.	This study showed a pipeline that combines transfer-learning feature extraction. Therefore, I can conclude from their paper that applying meta-heuristic optimization for feature selection and classifier tuning together with an improved SMOTE balancing step can dramatically boost pothole vs plain-road classification accuracy.
5	Real-time scene reconstruction and triangle mesh generation using multiple RGB-D cameras	2017	Point cloud smoothing, Mesh zippering, Direct triangle mesh generation, Mesh merging.	They proposed a real-time 3D reconstruction method with a GPU-optimized mesh merging method. They demonstrate both high mesh quality and significant acceleration compared to CPU-based methods. I can conclude that their GPU-parallel mesh approach, especially the MLS point cloud smoothing and mesh merging, gives insight into achieving fast, reliable surface modeling—knowledge that is valuable for enhancing the efficiency of mesh processing in pothole detection systems.

PROPOSED SOLUTION

- 1. Low-Cost Vision-Based Detection:** A monocular camera-based pothole detection system using deep learning eliminates the need for expensive LiDAR hardware, offering a lightweight, cost-effective alternative suitable for mass-scale autonomous vehicle deployment.
- 2. Robust Detection in Adverse Conditions:** By relying solely on visual data and monocular depth estimation, the system detects surface anomalies and water-filled potholes independent of laser reflectivity, while the compact camera form factor preserves vehicle aerodynamics without structural compromise.

PROPOSED SYSTEM

The system processes live video input from an onboard monocular camera and converts it into a 3D wireframe model for pothole detection and analysis.

1. Input Capture

Continuous RGB frames from the monocular camera mounted on the vehicle.

2. Object Detection & Segmentation

YOLOv8 detects potholes, cracks, and bumps with segmentation masks.

3. Depth Estimation

MiDaS predicts dense depth maps for each frame.

4. 3D Point Cloud Formation

The depth map + segmentation mask is converted into a localized point cloud.

5. Wireframe & Mesh Reconstruction

Open3D generates a surface mesh for shape and structure analysis.

6. Pothole Geometry Analysis

Depth, width, perimeter, and volume are computed using mesh features.

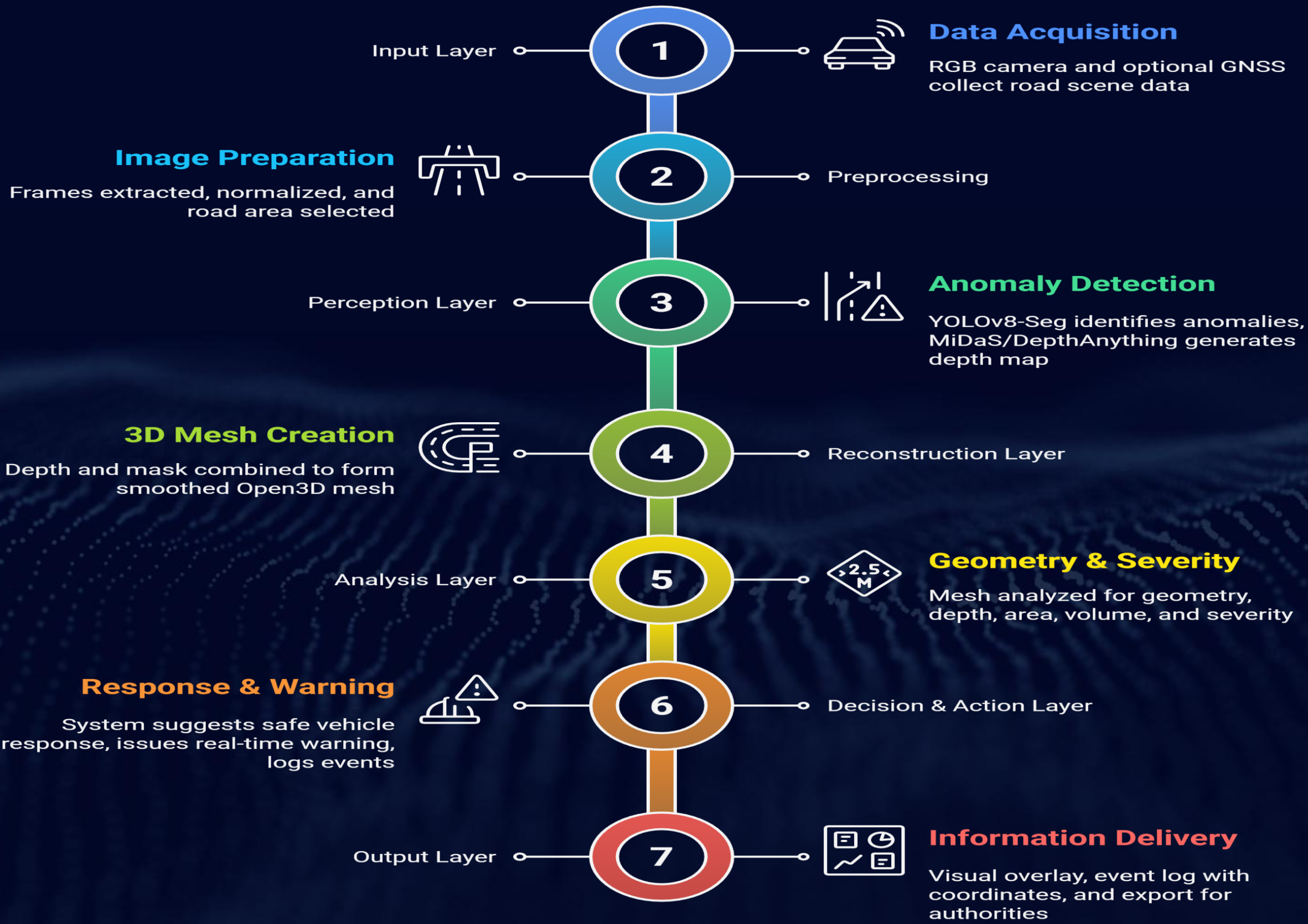
7. Decision Engine

Provides real-time action: slow down, steering adjustment, or event logging.

Purpose

To achieve **accurate, low-cost, real-time** pothole detection suitable for autonomous vehicles and road maintenance authorities.

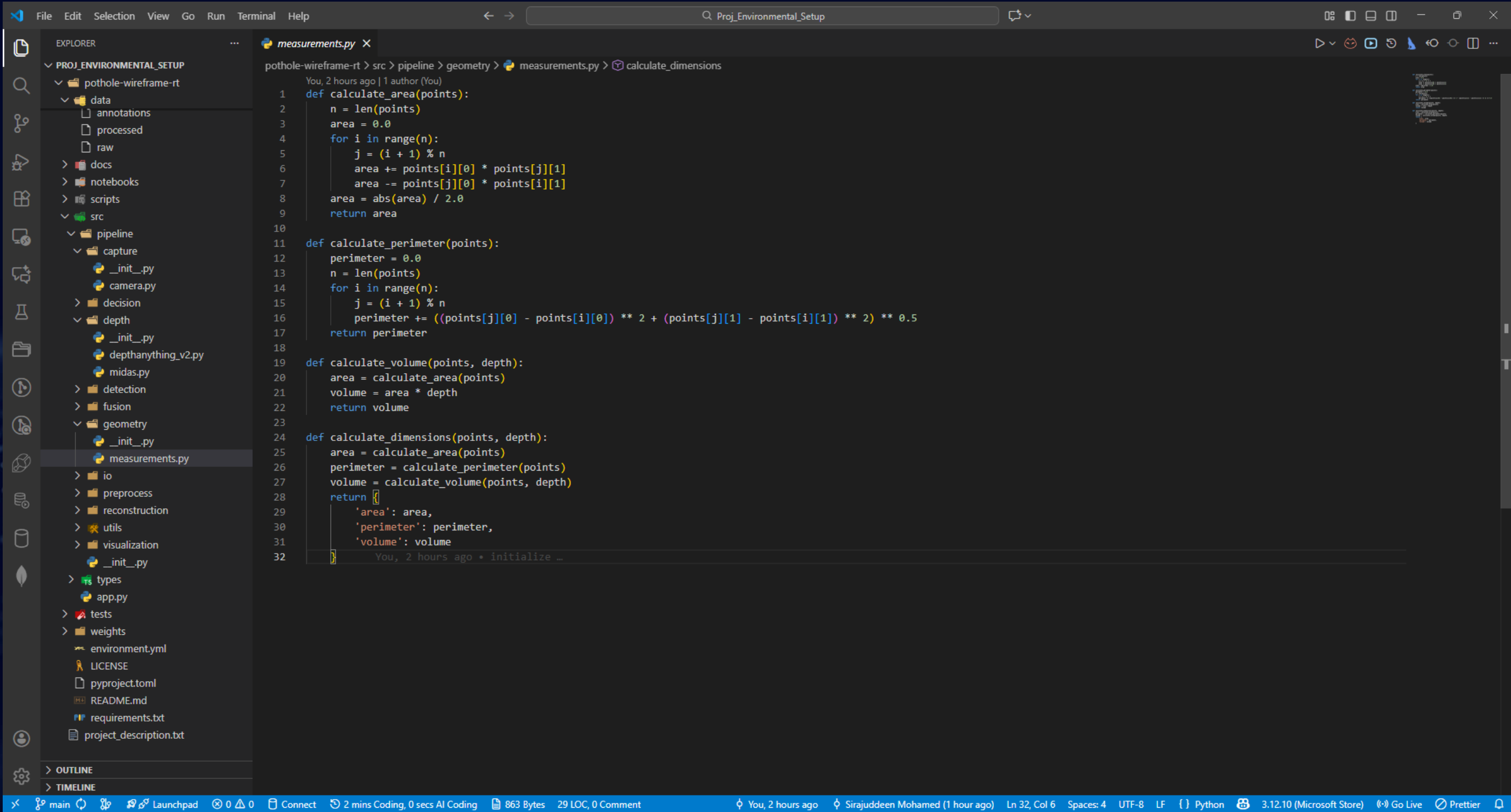
PROPOSED ARCHITECTURE



SYSTEM REQUIREMENT

Component	Specification	Software Requirements
Camera	Monocular RGB camera, 1080p@30–60 FPS, wide-angle 120° FOV	Programming Language: Python 3.10+
Processing Unit	NVIDIA Jetson Nano / Xavier NX or Laptop GPU (RTX 3060+)	Deep Learning Framework: PyTorch
CPU	Quad-core 2.0 GHz or above	ML Models: YOLOv8-Seg, MiDaS / DepthAnything V2
RAM	Minimum 8 GB (Recommended 16 GB)	3D Toolkit: Open3D
Storage	256 GB SSD for datasets + logs	Development Tools: Anaconda / Virtual Environment Jupyter Notebook / VS Code CUDA Toolkit (if using NVIDIA GPU) CARLA / Gazebo Simulation Docker
Optional Sensors	GNSS module for mapping pothole locations	OS: Ubuntu 20.04/22.04 or Windows 10/11

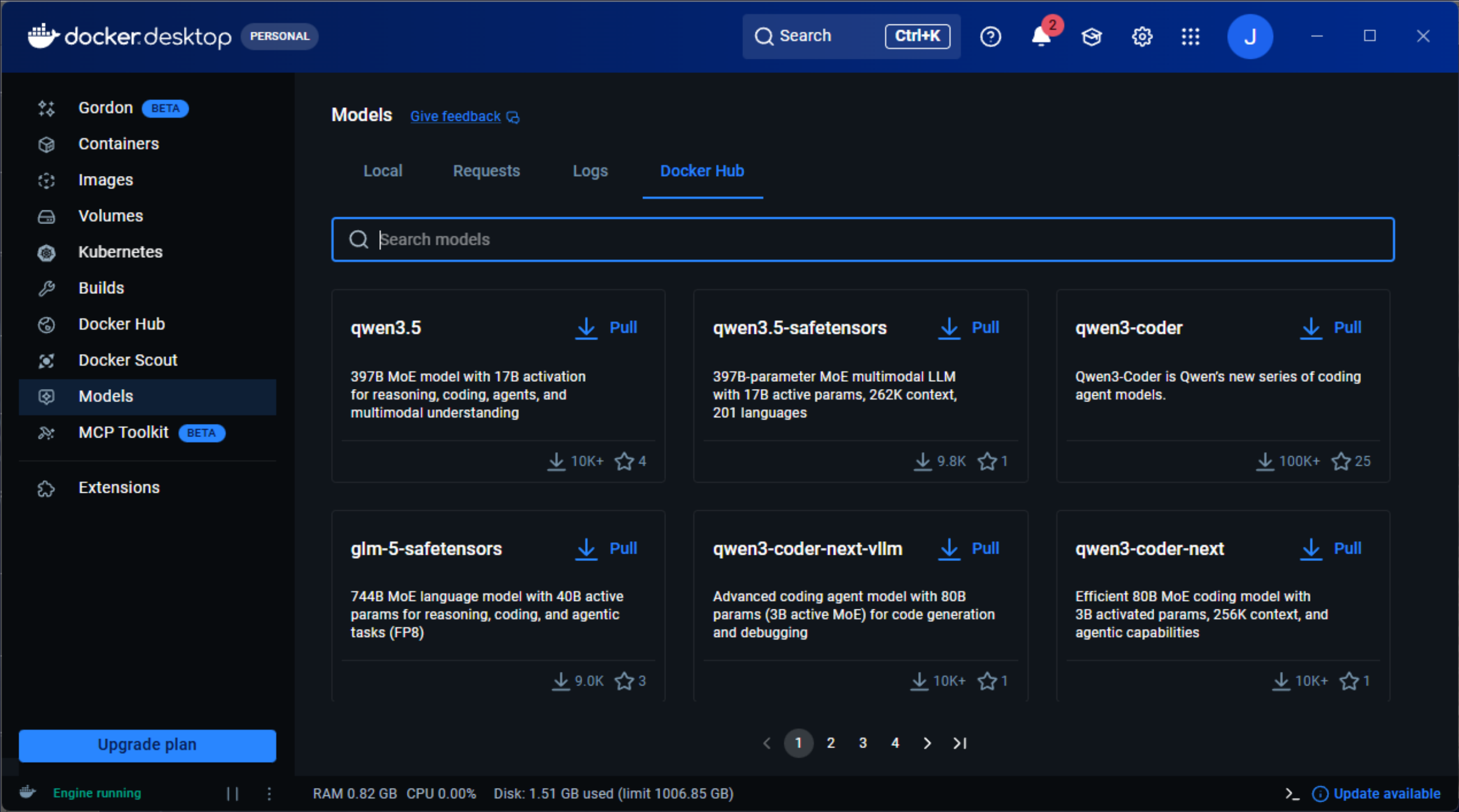
ENVIRONMENTAL SETUP



ENVIRONMENTAL SETUP

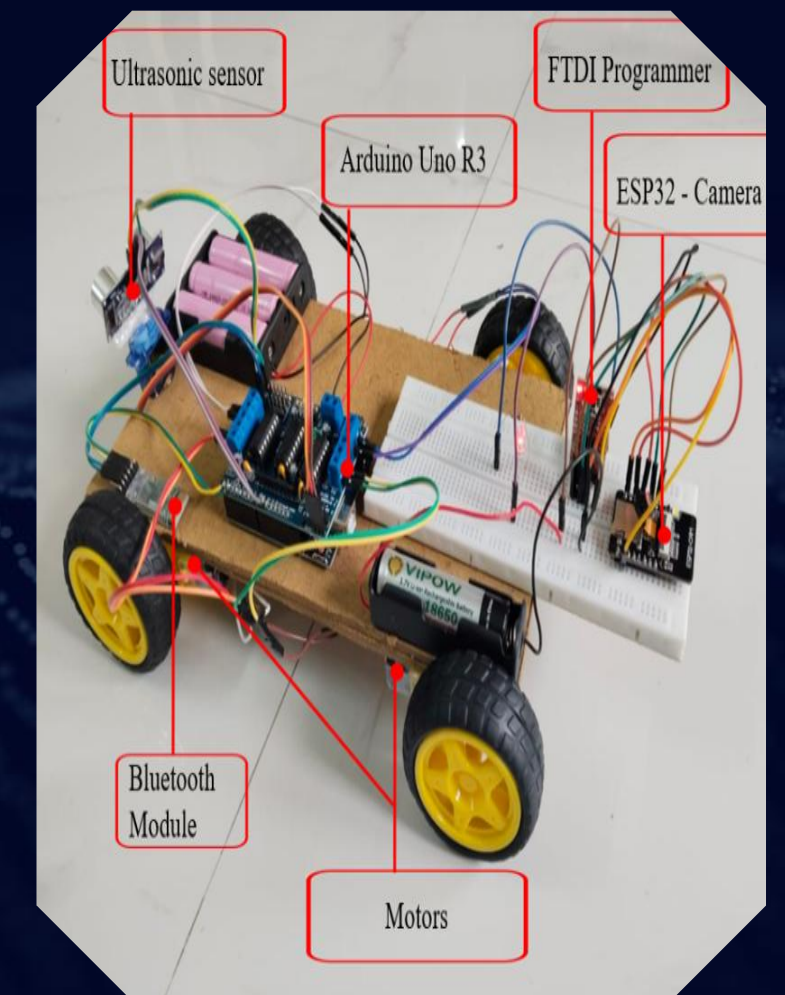


ENVIRONMENTAL SETUP



IMPLEMENTATION DETAILS

- ❖ Development environment configured: Python 3.10+, Anaconda, CUDA Toolkit, Docker and VS Code set up on an RTX 3070 Ti system. (21.75 TFLOPS)
- ❖ YOLOv8 object detection model installed and tested on pothole dataset images; model loads and runs inference successfully.
- ❖ Pothole detection implemented using bounding box method — potholes are identified and localized with rectangular bounding boxes per frame.
- ❖ MiDaS / DepthAnything V2 integrated as a standalone module; per-pixel depth maps are being generated from monocular RGB frames.
- ❖ Initial sensor integration completed: detection output (bounding boxes) and depth maps are being linked together in the pipeline.
- ❖ Detection accuracy is currently moderate — false positives observed under varying lighting and surface conditions; model tuning is ongoing.
- ❖ Wireframe-based 3D mesh generation from monocular depth is the next planned stage; design and research are in progress but not yet implemented.
- ❖ Transition from 2D bounding box detection to full 3D geometric wireframe representation is the core objective



RESULT & DISCUSSION

- ❖ YOLOv8 successfully detected potholes across test images; bounding boxes were generated around pothole regions in real time.
- ❖ Detection runs at acceptable inference speed on GPU; real-time processing feasibility on a single camera feed has been confirmed.
- ❖ Depth maps generated by MiDaS showed visible depth contrast between road surface and pothole regions, confirming monocular depth estimation is viable for this use case.
- ❖ Initial integration of detection output and depth maps established; bounding box coordinates are being correlated with corresponding depth regions.
- ❖ Detection accuracy is currently moderate — the model shows occasional false positives under challenging lighting, wet surfaces, and shadow conditions.
- ❖ The current bounding box approach lacks geometric depth information, which limits severity assessment and precise boundary localization of potholes.
- ❖ These limitations directly motivate the proposed transition to a wireframe-based 3D mesh pipeline using monocular depth.



CONCLUSION & FUTURE ENHANCEMENT

The current demonstration is a fully functional, LiDAR-free pothole detection pipeline using a single RGB camera, YOLOv8-Seg, MiDaS depth estimation, Open3D wireframe reconstruction, and a geometry-aware decision engine. The end-to-end system detects potholes, generates 3D wireframe meshes, computes geometric severity metrics (depth, area, perimeter), and triggers proactive vehicle responses in real time — all without expensive sensors or calibration overhead.

Future Plan: Temporal modeling (RNN/Video Transformers) for frame-to-frame wireframe stability | Metric depth via camera calibration for accurate pothole volume | Multi-defect detection (cracks, bumps) | Vehicle dynamics integration for adaptive deceleration | Crowdsourced road monitoring | Edge/embedded optimization for deployment on NVIDIA Jetson platforms.



CONCLUSION

REFERENCE

- [1] Deep Learning Enhanced Feature Extraction
- [2] An Enhanced YOLOv8 Model for Real-Time and Accurate Pothole Detection and Measurement
- [3] Robust Video Based Pothole Detection and Area Estimation for Intelligent Vehicles with Depth Map and Kalman Smoothing
- [4] Mobile3DRecon: Real-time Monocular 3D Reconstruction on a Mobile Phone
- [5] Pothole and Plain Road Classification Using Adaptive Mutation Dipper Throated Optimization and Transfer Learning for Self-Driving Cars.
- [6] Real-time scene reconstruction and triangle mesh generation using multiple RGB-D cameras



THANK
YOU